INCH-POUND

MIL-DTL-16377/12B(SH)
2 August 1996
SUPERSEDING
MIL-F-16377/12A(SH)
18 March 1982

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT, GENERAL LIGHTING, 51 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 333.1, 333.2, 341.1, 341.2, 342.2 AND 342.3

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-16377 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

All dimensions in inches.
For tolerances, other than specified herein, see 3.5.2 of MIL-DTL-16377.

AMSC N/A FSC 6210
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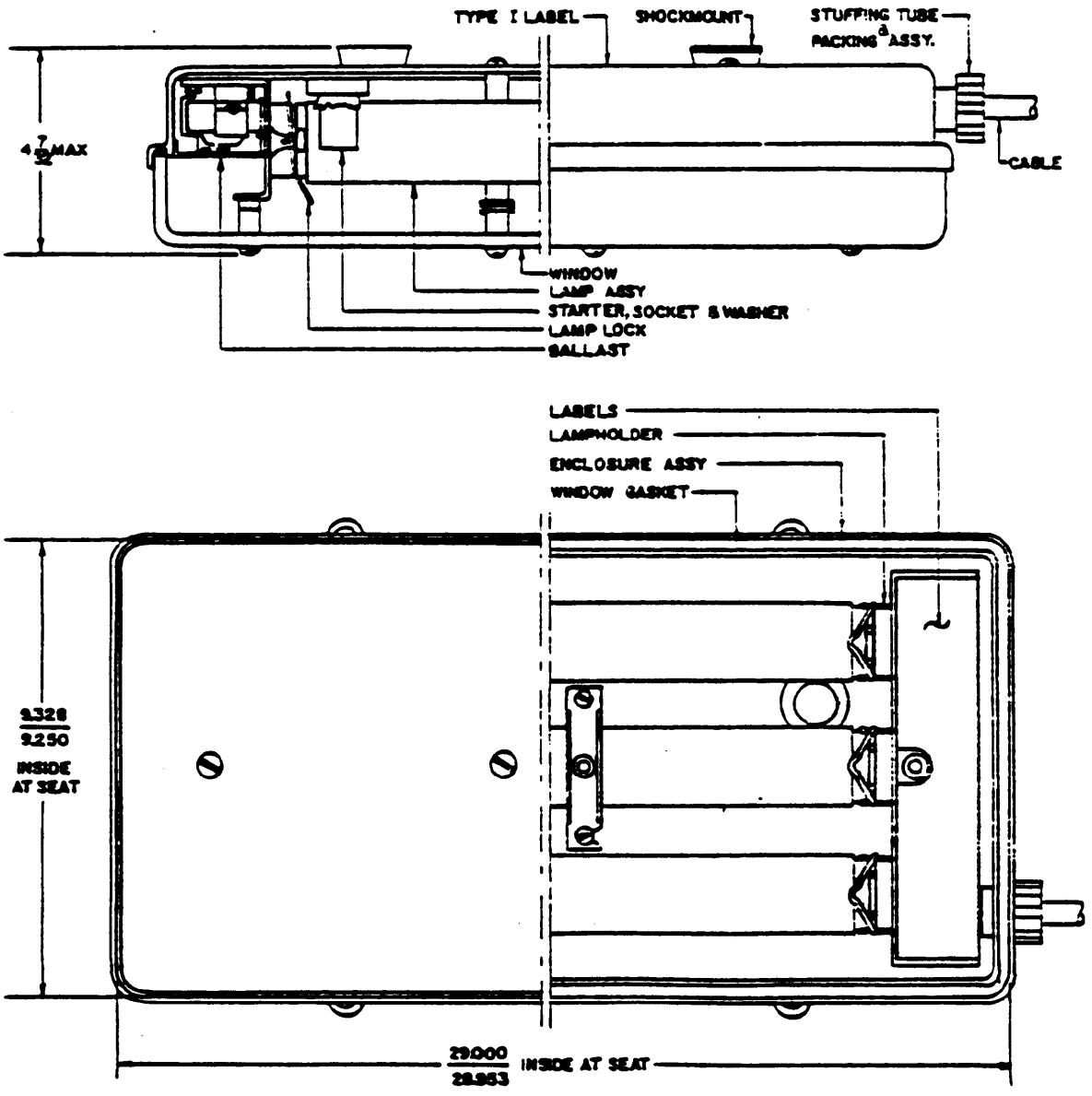


FIGURE 1. Typical view of Symbols 331.1 and 342.2.

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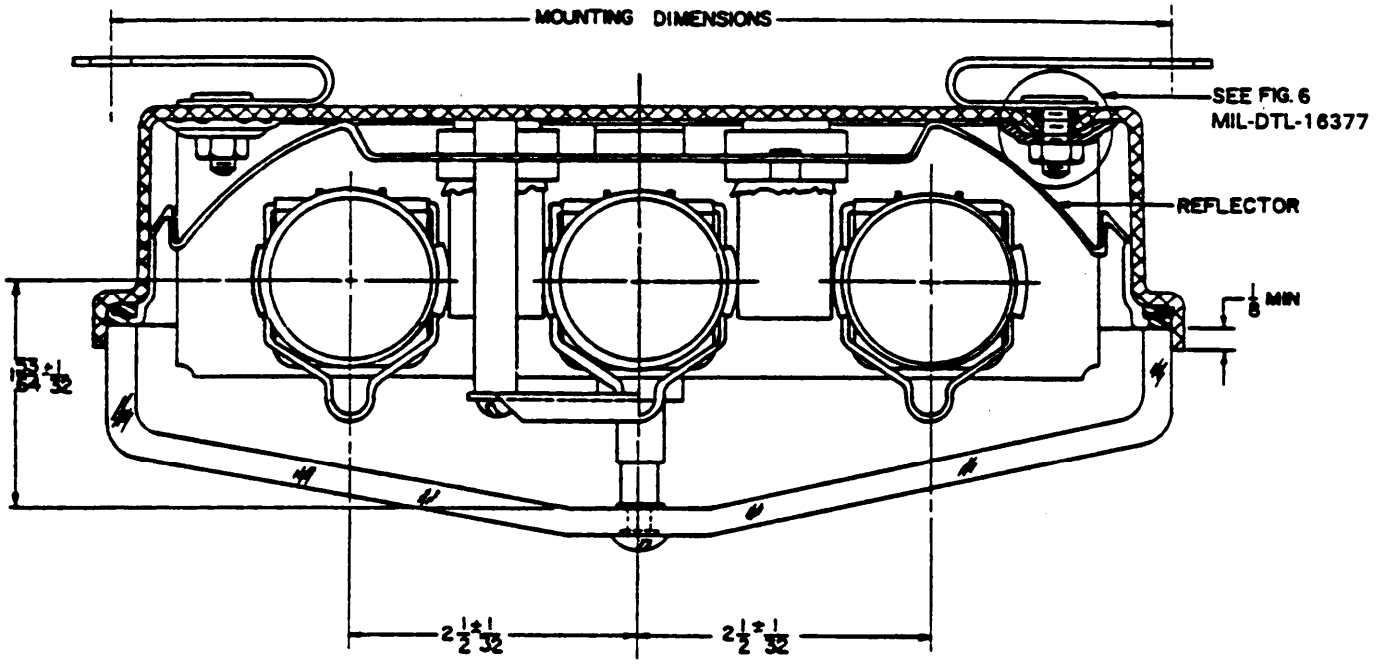


FIGURE 2. Typical section for Symbols 333.1 and 342.2

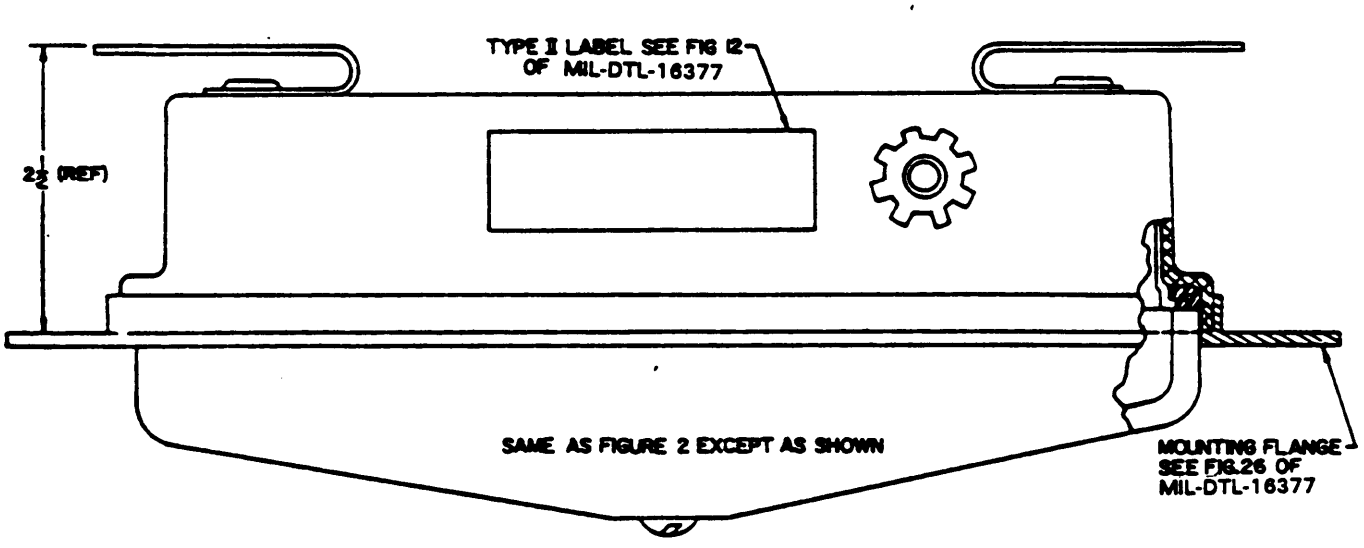


FIGURE 3. Typical section for Symbols 333.2 and 342.3.

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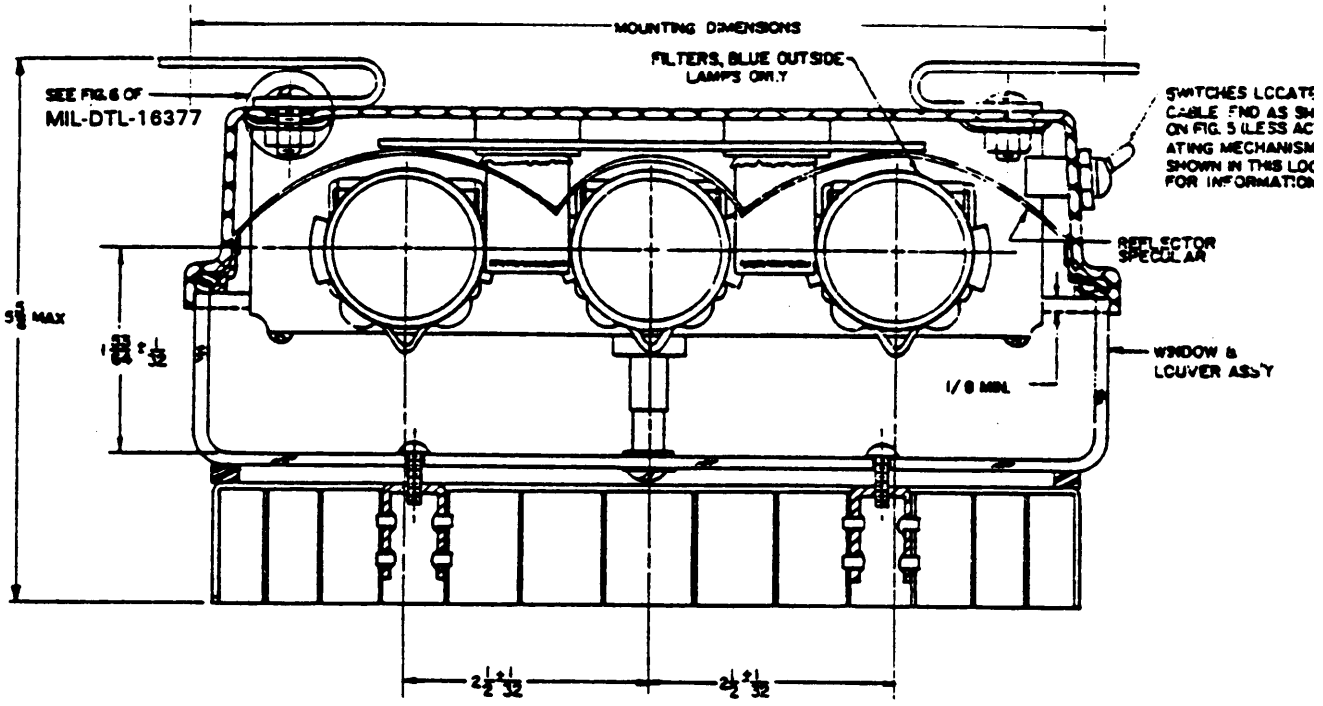


FIGURE 4. Typical section for Symbol 341.1.

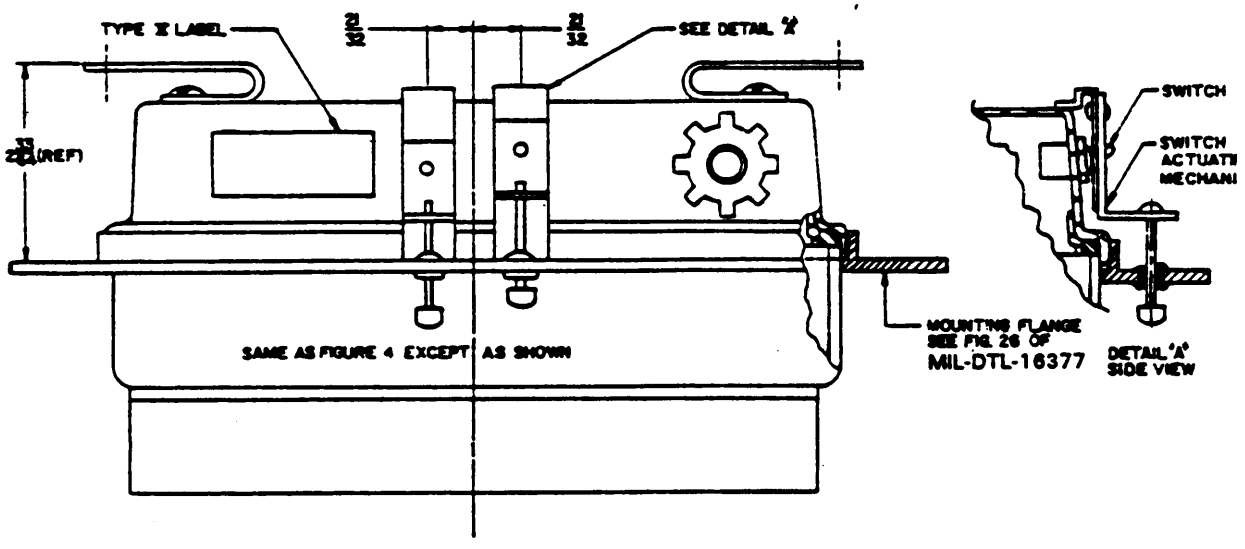


FIGURE 5. Typical section for Symbol 341.2.

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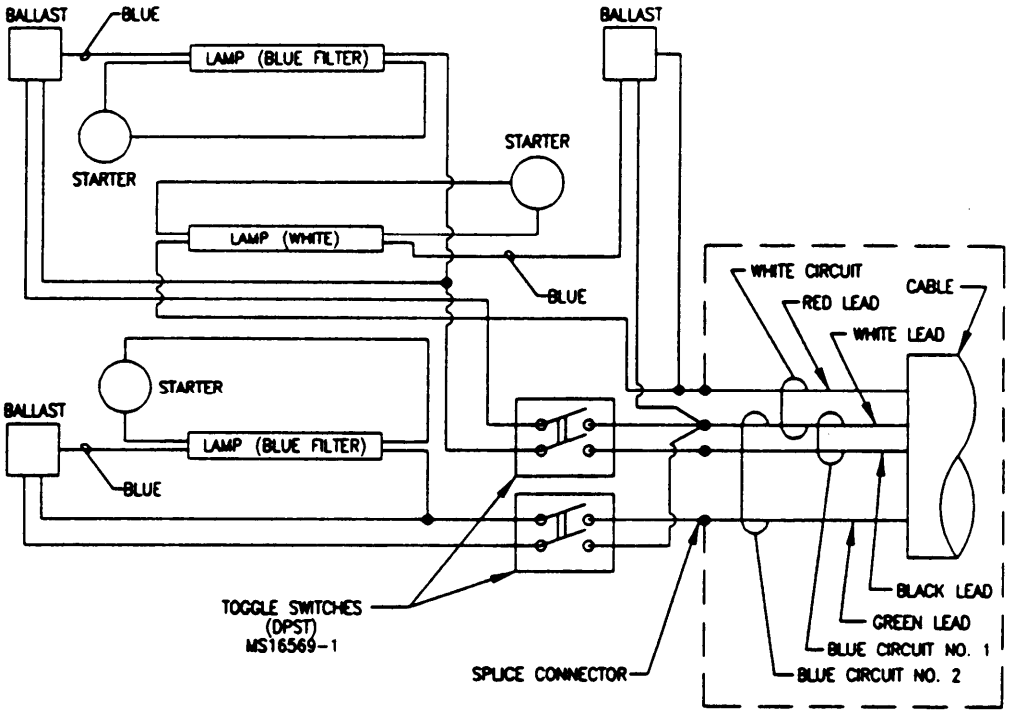
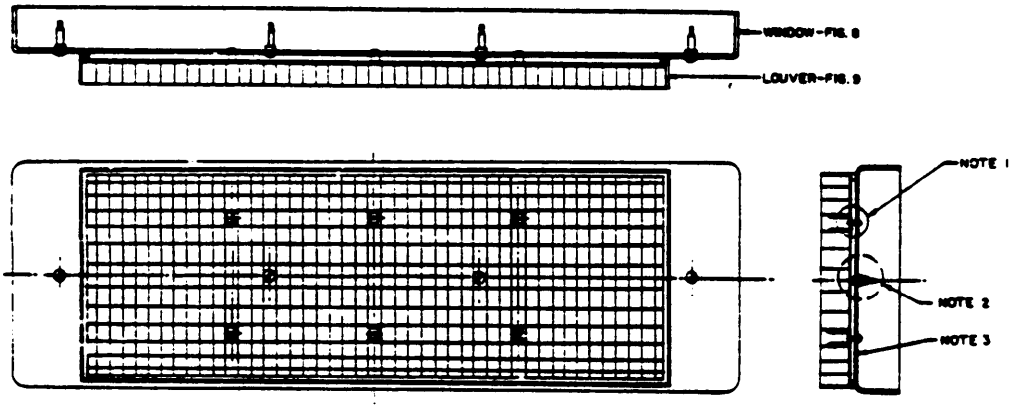


FIGURE 6. Wiring diagram for Symbols 341.1 and 341.2

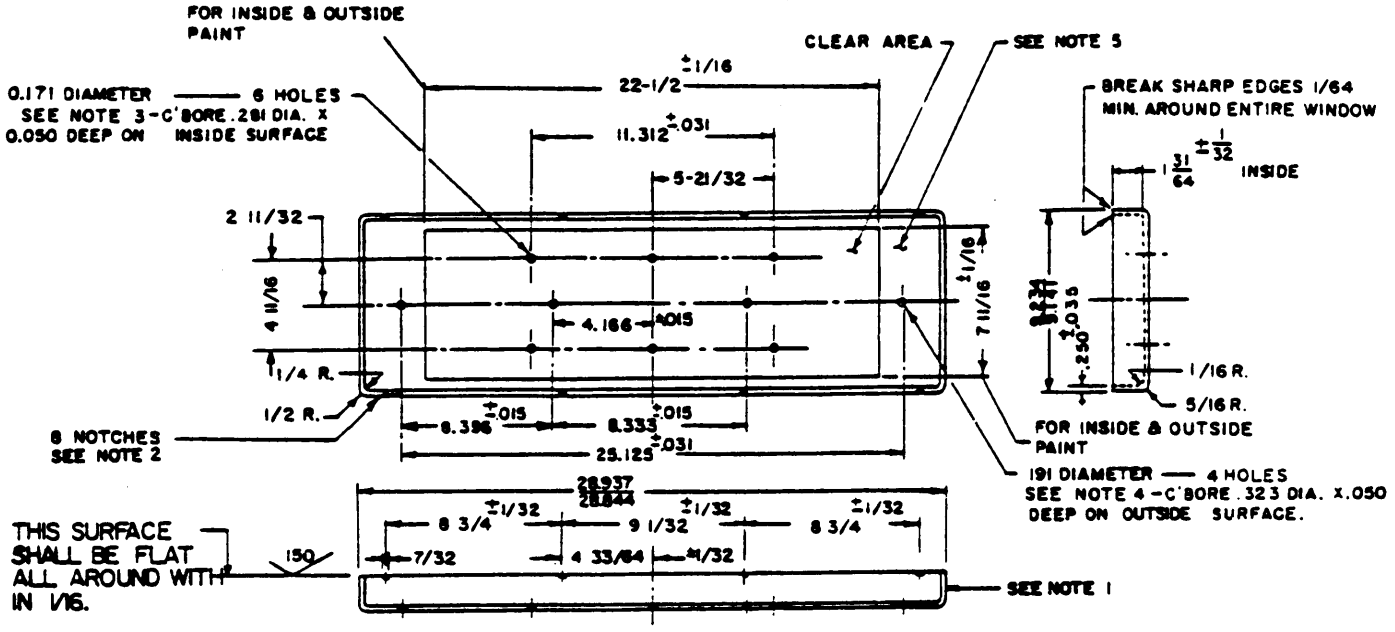


NOTES:

1. Attachment of louver to window shall be in accordance with 3.7.4.2.2 of MIL-DTL-16366.
2. Four short securing screw assemblies shall be installed in accordance with 3.7.4.3 of MIL-DTL-16377.
3. When required, one type VI label shall be applied in accordance with 3.7.4.3 of MIL-DTL-16377.

FIGURE 7. Window assembly, clear black shielded (CIC) for Symbols 341.1 and 341.2.

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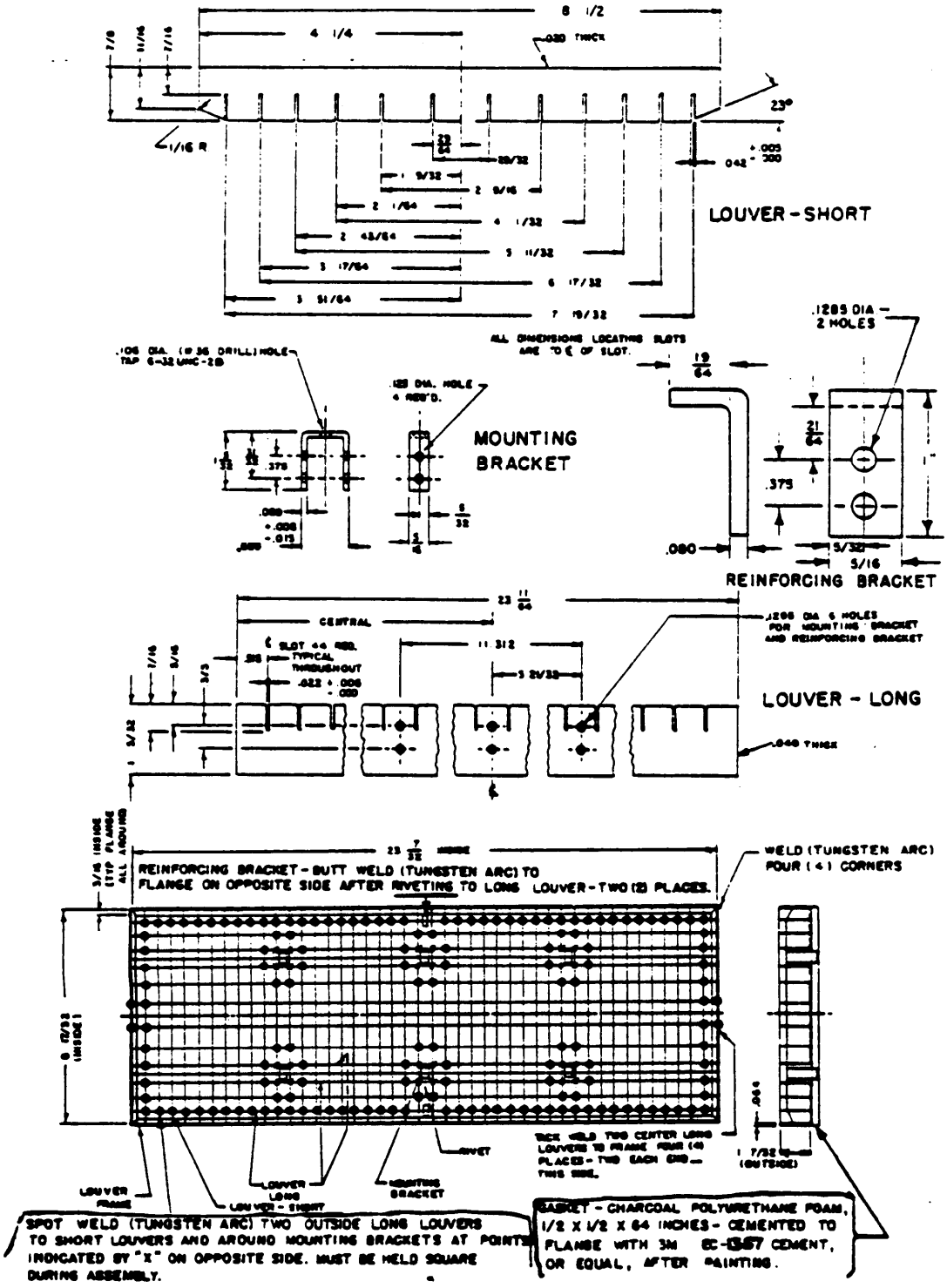


NOTES:

1. Material shall be 1/4 inch thick, clear transparent acrylic plastic, type CLEAR-1 of MIL-P-24191.
2. When required, notches shall be filed as shown in accordance with 3.7.4.4 of MIL-DTL-16377.
3. Break sharp edges on the outside surface of all holes.
4. Break sharp edges on the inside surface of all holes.
5. The inside surface of the window except the area indicated as “CLEAR AREA” shall be finished black. The outside surface of the window except the area indicated as “CLEAR AREA” shall be finished black. Finishing shall be in accordance with 3.8.4.1 of MIL-DTL-16377.

FIGURE 8. Window details.

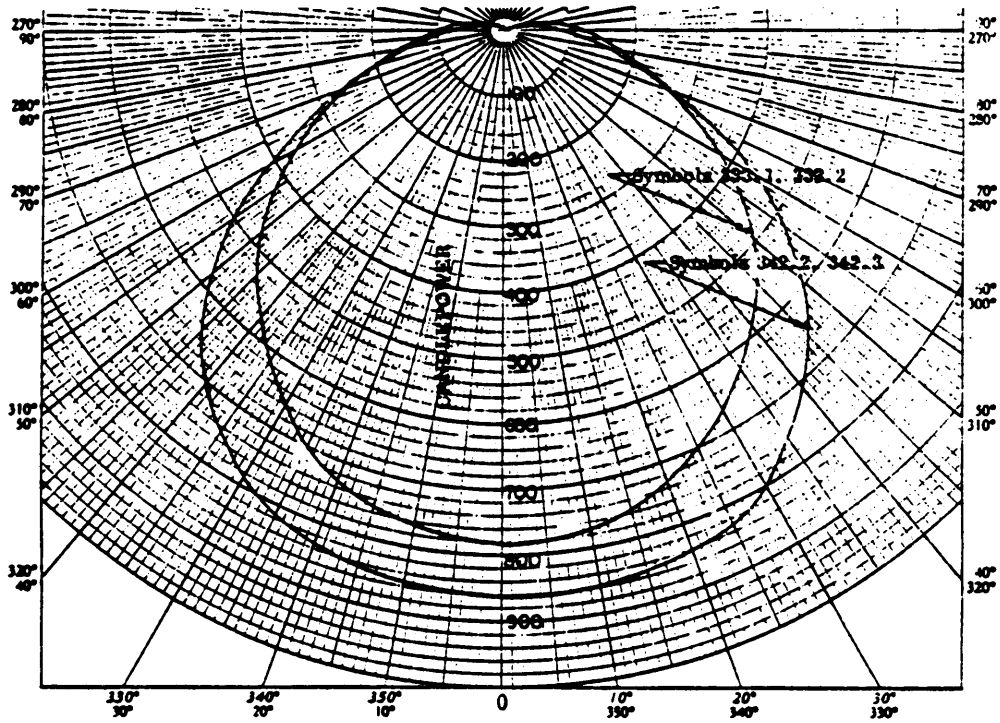
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NOTE: Louver assembly shall be finished black in accordance with 3.8.4.2 or MIL-DTL-16377.

FIGURE 9. Louver assembly.

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ANGLE	CANDLEPOWER SYMBOL NUMBER	
	333.1, 333.2	342.2, 342.3
0	780	850
5	770	850
10	760	840
15	750	840
20	730	825
25	700	820
30	660	790
35	620	750
40	570	700
45	530	650
50	475	570
55	430	550
60	370	410

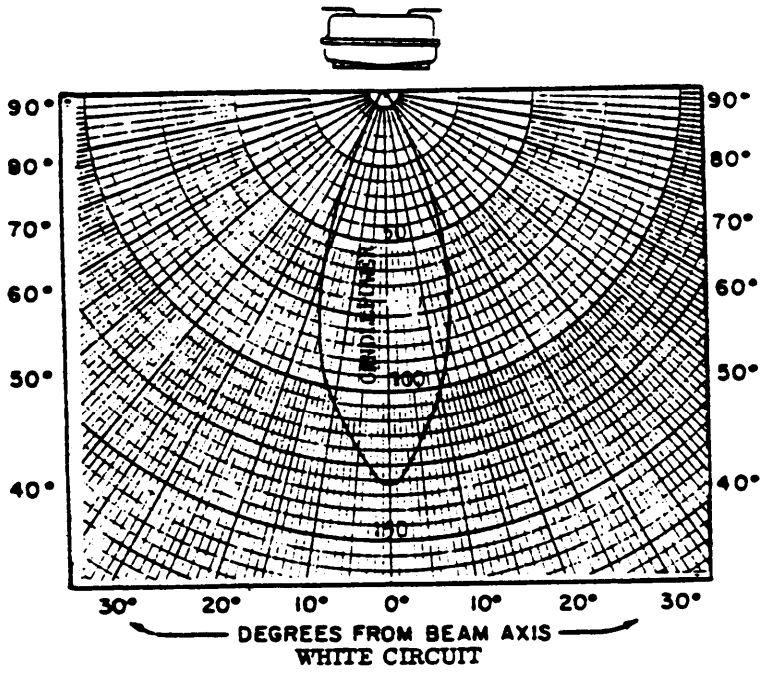
ANGLE	CANDLEPOWER SYMBOL NUMBER	
	333.1, 333.2	342.2, 342.3
65	310	320
70	260	250
75	210	200
80	170	150
85	130	110
90	90	100
95	70	50
100	40	40
105	30	20
110	25	0
115	10	0
120	0	0
125	0	0

Photometric test data:

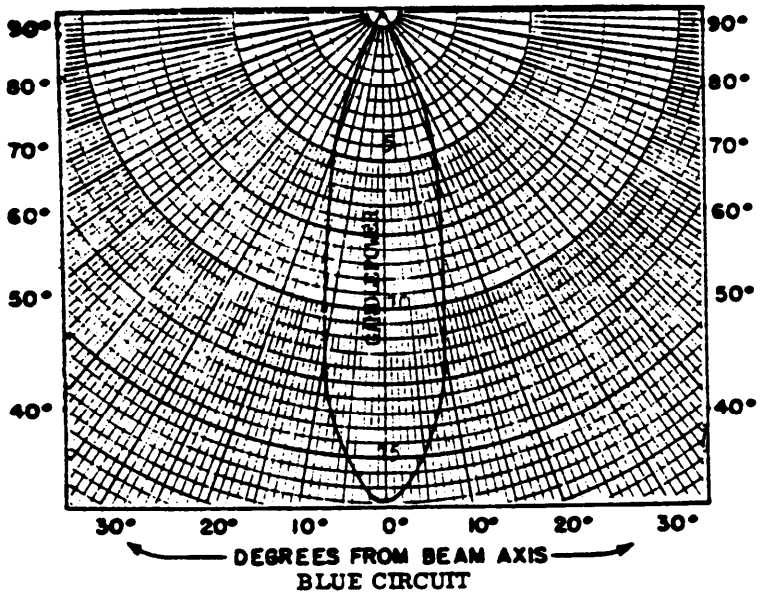
Distance: 10 feet.
Plane: At right angles to long axis of fixture through geometric center of lamp.
Voltage: 115 volts, 60 Hz.
Ambient temperature: 76°F
Lamps: Use lamps furnished with fixtures seasoned for at least 100 hours.
Ballast: Use ballast furnished with fixture.
Lamp correction: Distribution curves based on 1280 lumens per lamp.

FIGURE 10. Candlepower distribution curves for Symbol 333.1, 333.2, 342.2, and 342.3

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Tests shall be made with center daylight lamp energized and blue filters in place on outside de-energized lamps.



Tests shall be made with center daylight lamp de-energized and blue filters in place on outside energized lamps.

Photometric test data:

Distance: 10 feet.
Plane: At right angles to long axis of fixture through geometric center of lamps.
Voltage: 115 volts, 60 Hz, 1 phase.
Ambient temperature: 76°F.
Lamps: Use daylight lamps furnished with fixture.
Ballasts: Use ballast furnished with fixture.
Correction: Distribution curves based on 1075 lumens per lamp.

FIGURE 11. Candlepower distribution curves for Symbols 341.1 and 341.2.

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REQUIREMENTS:

CLASSIFICATION:
Type: I
Class: 2

SYMBOL NUMBERS:
333.1: Complete lighting fixture assembly, standard mounting, with three 17-watt cool white fluorescent lamps and white diffusing window assembly.
333.2: Complete lighting fixture assembly, flush mounting with three 17-watt cool white fluorescent lamps and white diffusing window assembly.
341.1: Complete lighting fixture, standard mounting, blue and white (Command Information Center (CIC) Lighting), clear black shielded with three 20 watt daylight lamps with blue filters over each outside lamp.
341.2: Complete lighting fixture, flush mounting blue and white (Command Information Center (CIC) Lighting), clear black shielded, with three 20 watt daylight lamps with blue filters over each outside lamp.
342.2: Complete lighting fixture assembly, standard mounting with three 17-watt cool white fluorescent lamps and clear and white prismatic window assembly.
342.3: Complete lighting fixture assembly, flush mounting, with three 17-watt cool white fluorescent lamps and clear and white prismatic window assembly.

DESIGN:
Electrical rating: 51 watts, 120 volts, 60 Hertz.
Dimension and configurations: See figures 1, 2, 3, 4, and 5 and figure 10 of MIL-DTL-16377.
Photometric data: See figures 10 and 11.
Lampholder spacing: See figure 21 of MIL-DTL-16377.
Wiring diagram: See figure 24 of MIL-DTL-16377 for symbols 333.1, 333.2, 342.2, and 342.3 and figure 6 for symbols 341.1 and 341.2.
Mounting dimensions:
Symbols 333.1, 341.1 and 342.2: See figure 25 of MIL-DTL-16377.
Symbols 333.2, 341.2 and 342.3: See figure 26 of MIL-DTL-16377.
Window assembly:
Symbols 333.1 and 333.2: Military part number M16377/12-001, figure 27 of MIL-DTL-16377.
Symbols 341.1 and 341.2: Military part number M16377/12-003, see figure 7.
Symbols 342.2 and 342.3: Military part number M16377/12-004, figure 27 of MIL-DTL-16377.

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Enclosure: See 3.5.11 of MIL-DTL-16377. Watertight for 333.1, 333.2, 342.2, and 242.3. Dripproof (15°) for 341.1 and 341.2. Boot for toggle switch: Neoprene rubber with integral hexagon nut, enclosed or through type in accordance with MIL-B-5423. Finishing for enclosures for Symbols 341.1 and 341.2: Exterior surfaces shall be finished with black baking enamel.

Cable: 10 feet of type LSTHOF-3, in accordance with MIL-C-24643/3 three conductors (white, black and red) for Symbols 333.1, 333.2, 342.2, and 342.3 and LSFHOF-3 for Symbols 341.1 and 341.2.

Weight:

Symbols 333.1 and 342.2: 15 lbs 11 oz. maximum.
Symbols 333.2 and 342.3: 17 lbs 14 oz. maximum.
Symbol 341.1: 19 lbs 4 oz. maximum.
Symbol 341.2: 21 lbs 7 oz. maximum.

Designation and marking: The following labels shall be furnished in accordance with MIL-DTL-16377 applied as shown on figures 1 and 3:

(a) Two type I.
(b) One type II for Symbols 333.2 and 342.3 only.
(c) One type IV. The information to be shown on the type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY." List all standard parts shown in table I and window assemblies of table IV by name and stock number only.
(d) For Symbol 341.1 only: Legend on the protective cover as shown on figure 19 of MIL-DTL-16377, shall have the cable connection portion covered with a type III label with wiring diagram as shown by dotted box on figure 6.
(e) For Symbol 341.2 only: A type III label with wiring diagram as shown by dotted box on figure 6 shall be applied as shown on figure 5.

Standard parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

QUALITY ASSURANCE: Quality assurance shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:
Complete fixtures: See table II.
Window assemblies: See table III.
Conformance: Same as the first article inspection.
Inspection of product for delivery:
Examination: See MIL-DTL-16377.
Examination for preparation of delivery: See MIL-DTL-16377.
Conformance tests: See MIL-DTL-16377.

GENERAL INFORMATION:

Assignment of Military part numbers and general information is shown in table IV.

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TABLE I. Standard parts

Name	Specification or Military part number	National stock number (NSN)	Remarks
Lamp	1/ A-A-50597		For Symbols 333.1 333.2, 342.2 and 342.3
Lamp	2/ W-L-116/18	6240-00-152-2993	For symbols 341.1 and 341.2
Lamp lock	M16377/40-002	6250-00-399-7040	
Lampholder	UL-542	6250-00-284-0481	Supplement SA, Figure SA2.2
Startersocket	UL-542	6250-00-295-2738	Supplement SA, Figure SA2.4
Starter	20W, type III of W-S-755	6250-00-299-2884	Commercial designation: FS2
Ballast	M16377/72-001		
Shockmount	M16377/41-009 or M16377/41-010 or M16377/41-008	6210-00-973-1747 6210-00-973-1748	See 3.5.16 of MIL-DTL-16377
Red filter tube	M16377/42-003	6210-00-295-2838	Not furnished. See 3.7.7 of MIL-DTL-16377
Blue filter tube	M16377/42-005	6240-00-441-5138	Furnished with Symbols 341.1 and 341.2 only
Switch, toggle DPST	MS16569-1	5930-00-719-9659	For Symbols 341.1 and 341.2 only

1/ Industry number F17T12/CW.
2/ Industry number F20T12/D.

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TABLE II. First article and comparison inspection for complete fixtures.

Test	Requirement	Test method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14.4	1/
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	1/
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Noise	3.5.14	4.8.9	
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	

1/ Watertight for Symbols 333.1, 333.2, 342.2, and 342.3.
Dripproof 15 degrees for Symbols 341.1 and 341.2.

TABLE III. First article and comparison inspection for window assemblies.

Test	Requirement	Test method	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	1/
Light output	3.7.1	4.8.6.1	Candlepower

See footnote at top of next page.

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1/ Watertight for Symbols 333.1, 333.2, 342.2, and 342.3.
Dripproof 15 degrees for Symbols 341.1 and 341.2.

TABLE IV. Part number and general information.

Military part number 1/ M16377/12-	Item name	National stock number (NSN)	Remarks
333.1	Complete fixture, Symbol 333.1	6210-00-548-0222	2/
333.2	Complete fixture, Symbol 333.2		2/
341.1	Complete fixture, Symbol 341.1	6210-00-455-0068	2/
341.2	Complete fixture, Symbol 341.2		2/
342.2	Complete fixture, Symbol 342.3		New
342.3	Complete fixture, Symbol 342.3		New
001	Window assembly, white diffusing	6210-00-295-2849	3/
003	Window assembly, clear black shielded	6210-00-725-9030	4/
004	Window assembly, clear and white prismatic		New

1/ Military part numbers M16377/12-342, 342.1 and 002 have been superseded by Military part numbers M16377/12-342.2, 342.3, and M16377/67-005 respectively.
2/ Formerly covered on Drawing NAVSHIPS 9000-S6401-74458 and MS19107.
3/ Formerly covered on MS16891-36.
4/ Formerly covered on MS16891-68.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
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